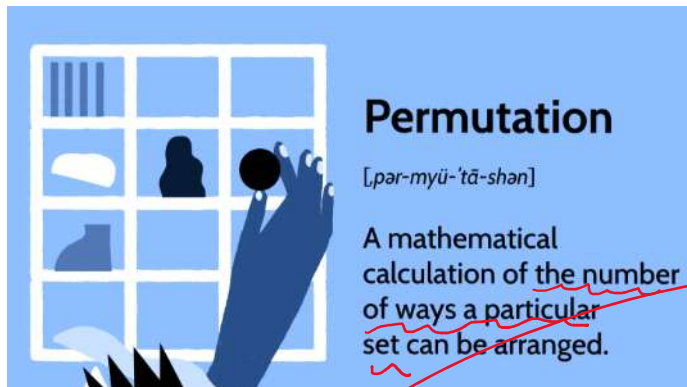


Permutations - Formula

Monday, May 5, 2025 9:44 AM



Formula

$${}_n P_r = \frac{n!}{(n-r)!}$$

n choose r

${}_n P_r$ = permutation = ${}_{11}P_3$
 n = total number of objects = 11
 r = number of objects selected = 3

E.g. let's say you need to create a 3-digit password (0...10).

$${}_{11}P_3 = \frac{n!}{(n-r)!} = \frac{11!}{8!}$$

Definition: Factorial

$$n! = n \times (n-1) \times (n-2) \times \dots \times 1$$

$$3! = 3 \times 2 \times 1$$

$$7! / 3! = 7 \times 6 \times 5 \times 4 \times 3! / 3! = 7 \times 6 \times 5 \times 4 = 840$$

$$\rightarrow 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$= \frac{11 \times 10 \times 9 \times 8!}{8!} = 990$$

If you create a 3-digit numeric password using the numbers from 0 to 10, what is the probability someone will guess that password? **There are 990 possible passwords; you choose 1 of those. So, the probability someone will guess your password is $1/990 = 0.001 = 0.001 = 0.001 = 0.1\%$**

$$1 \div 990 =$$

0.001010101010101

$1/990$ is approximately 0.01%

Permutations

Monday, March 10, 2025 12:58 PM

PERMUTATIONS

In the last lesson, you counted the possible ways that four children could line up for a photograph. A way was considered different if the children were in a different order. When order matters, you are counting the number of permutations.

In real life, the word combination is often used when really it is a permutation. When you use a combination lock, you have to enter the numbers in a certain order. A combination lock should really be called a permutation lock.

You will learn more about combinations in the next lesson. For now, remember that in a permutation problem, the order of the events or items matters.

Permutation lock

10 - 15 - 30



COUNTING PERMUTATIONS

Example In how many different ways can a student arrange five different books on a shelf?

Step 1

Count the number of choices the student has for each space on the shelf.

There are 5 choices for the first book. After a book is placed in that position, there are 4 choices for the second book. After each book is placed, there is 1 fewer choice. Notice how the choices count down after each book is placed.

Step 2

Multiply the choices: $5 \times 4 \times 3 \times 2 \times 1 = 120$

There are 120 different ways the books can be arranged on the shelf.

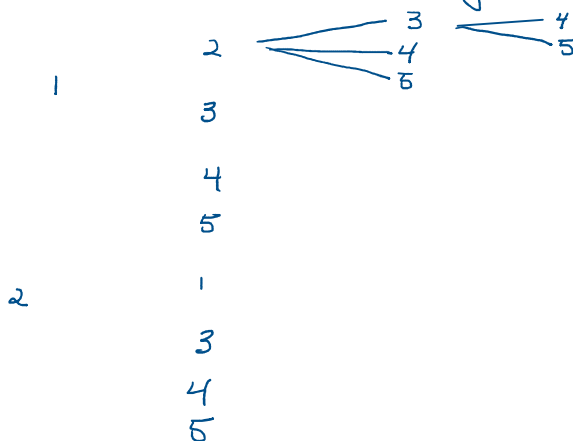


Recall the 4 children in the family photo >

$4 \times 3 \times 2 \times 1 = 24$ ways
OR

$$P(4,4) = \frac{4!}{0!} = 24$$

If we created a probability tree, it would look like this:



$$\begin{aligned} & \text{OR} \\ & P(5,5) = \frac{5!}{0!} \\ & = 120 \end{aligned}$$

Permutations, cont'd

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To solve any permutation problem, simply multiply the choices. Study how this basic idea can be used to solve a more complex problem.

FINDING PERMUTATIONS OF PART OF A GROUP

Example Fifteen runners are in a race. How many different finishes can there be for 1st, 2nd, and 3rd place?

Step 1

Count the number of choices. There are 15 runners who could finish in 1st place, 14 choices for 2nd place, and 13 choices for 3rd place. Remember, once a runner finishes 1st, that runner cannot also finish in 2nd or 3rd place.



Step 2

Multiply the choices: $15 \times 14 \times 13 = 2,730$ ✓

There are 2,730 different ways the runners could finish 1st, 2nd, and 3rd.

Will the choices always count down? Not always.

Some choices can be reused

Example A bike combination lock has three dials, each with 10 numbers. To open the lock, you turn the dials. A number can be used more than once. How many permutations are possible?

There are 10 choices for each dial: $10 \times 10 \times 10 = 1,000$.



Examples

Monday, March 10, 2025 1:00 PM

Solve.

- ✓ 1. Jasmine has 8 songs in a playlist. The playlist is set to play songs randomly. If she listens to 3 songs, how many different orders are possible if a song may not be repeated?

$$8 \times 7 \times 6 = 336 \text{ ways}$$

$$\begin{aligned} \text{OR: } P(8, 3) &= \frac{8!}{(8-3)!} = \frac{8!}{5!} \\ &= \frac{8 \times 7 \times 6 \times 5!}{5!} \\ &= 336 \end{aligned}$$

- ✓ 2. A company assigns a code to each customer. The code has 2 letters followed by a single-digit number from 1 to 9. If the letters must be different, how many codes can the company make? (Hint: There are 26 letters in the alphabet.)

$$26 \times 25 \times 9 = 5850$$

$$\begin{aligned} \text{OR: letters } P(26, 2) &= \frac{26!}{24!} = 26 \times 25 \\ &= 650 \end{aligned}$$

then each of these 650 has 1 number
so, $650 \times 9 = 5850$

- ✓ 3. For summer reading, Grace must choose 4 books out of a reading list of 12 books. In how many different ways could Grace complete the assignment?

<u>1st choice</u>	<u>2nd choice</u>	<u>3rd choice</u>	<u>4th choice</u>
12 Books	11 Books	10 Books	9 Books

$$12 \times 11 \times 10 \times 9 = 11,880 \text{ ways}$$

$$\begin{aligned} \text{OR, } P(12, 4) &= \frac{12!}{8!} = 12 \times 11 \times 10 \times 9 \\ &= 11880 \end{aligned}$$

- ✓ 4. Five friends are performing a song for a talent show. They plan to stand in a line. In how many different orders could they stand?

$$5 \times 4 \times 3 \times 2 \times 1 = 120 \text{ ways}$$

$$\text{OR } P(5, 5) = \frac{5!}{0!} = 120$$